# Pharma QC Data Integrity Risk Assessment

## ALCOA+ Compliance Dashboard and Risk Analysis

**Tools Used**

Microsoft Excel
Pivot Table Analytics
ALCOA+ Compliance Framework

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**Focus: Pharma QC Operations & Digital Transformation**

# Business Problem

- Pharmaceutical QC laboratories rely heavily on **manual documentation processes**.
- Manual data entry increases risk of **missing data and undocumented corrections.**
- Such documentation issues can impact **ALCOA+ data integrity principles.**
- These risks can impact **regulatory inspection readiness.**

**Key message**
Manual QC documentation introduces measurable data integrity issues.

# Objective of Analysis

The objective of this analysis is to identify data integrity risks in QC documentation processes using data-driven analysis.

**Focus Areas**

- Missing data incidents
- Frequency of documentation corrections
- Review status distribution
- Analyst-level risk score

# Methodology

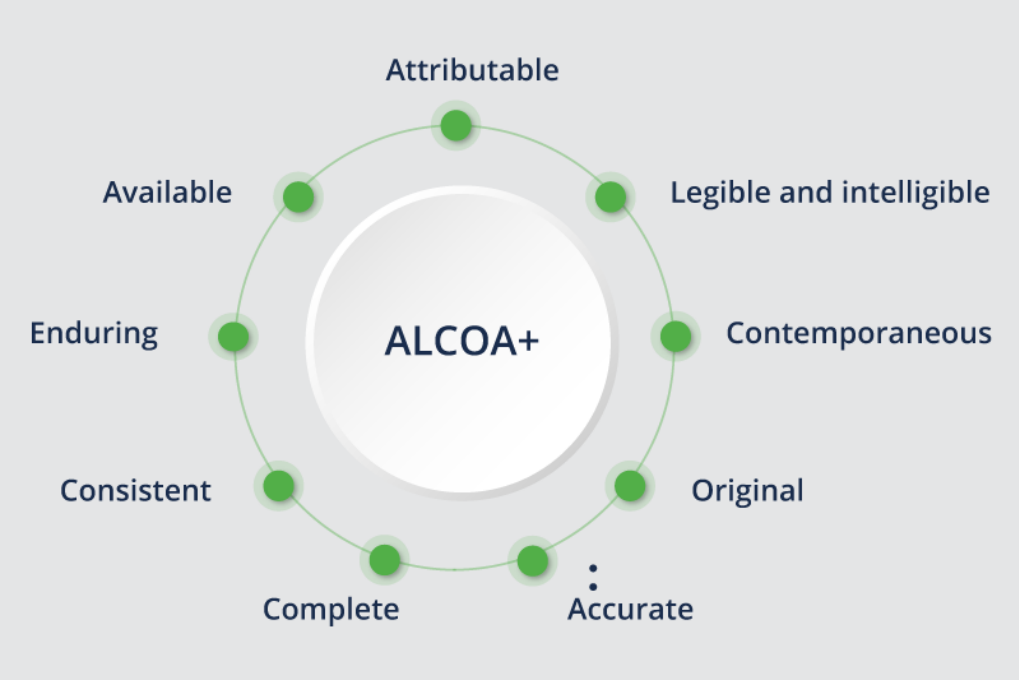
A simulated dataset of 200 QC laboratory records was analyzed using Excel-based analytics.

**Dataset Fields**

- Analyst
- Product
- Test Type
- Missing data
- Corrections Made
- Review Status

**Analytical Methods**

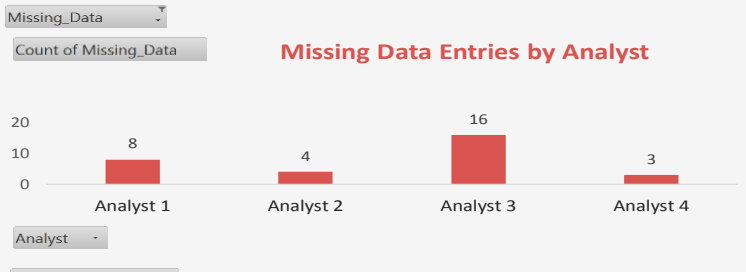
- Pivot table analysis
- Dashboard visualization
- Risk score calculation

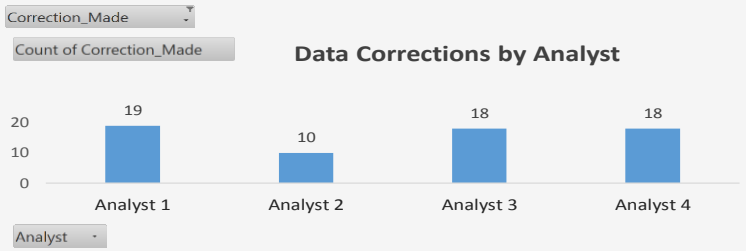


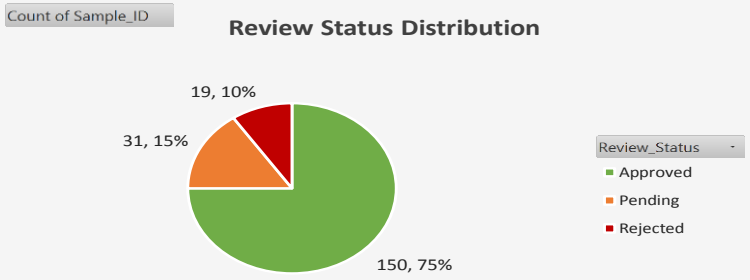
# Pharma QC Data Integrity Dashboard (ALCOA+ Compliance Monitoring)

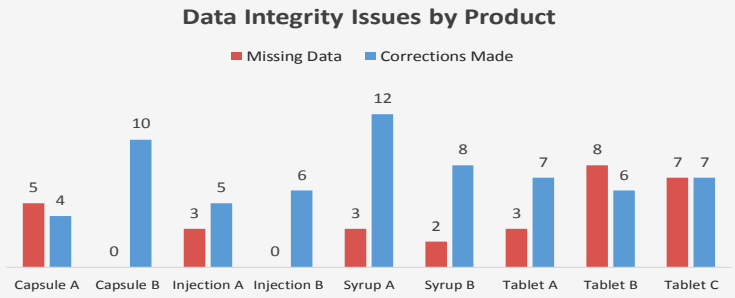
| Total Samples Tested | Data Integrity Issues | Missing Data Records |
|---|---|---|
| 200 | 87 | 31 |

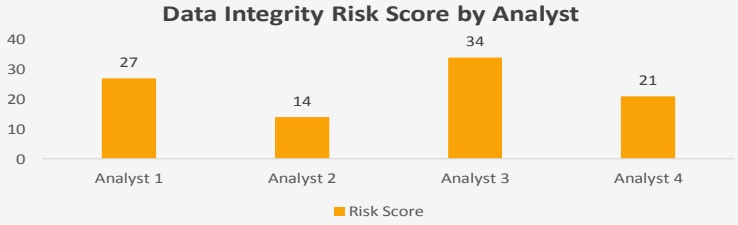
| Average Turnaround Time (Days) | Data Integrity Risk Rate |
|---|---|
| 3 | 44% |




Dashboard highlighting data integrity indicators including missing data frequency, corrections trends, review status distribution, and analyst-level risk scores.

**Key Takeaway**

Multiple documentation issues were observed across analysts and products.

# Dashboard Observations

**1. Significant Data Integrity Risk Identified**

Out of 200 QC records analyzed, a total of **87 data integrity issues** were identified, resulting in a **44% data integrity risk rate**. This indicates a notable level of documentation inconsistencies within QC laboratory records.

**2.Missing Data Indicates Documentation Gaps**

The dashboard shows **31 records with missing data**, suggesting incomplete documentation in QC testing records. Missing entries represent potential non-compliance with ALCOA+ completeness requirements.

**3.High Frequency of Documentation Corrections**

Manual corrections were observed across all analysts, indicating reliance on manual documentation processes. Frequent corrections increase the risk of data manipulation concerns during regulatory inspections.

**4.Uneven Distribution of Risk Across Analysts**

The data integrity risk score varies significantly across analysts, with **Analyst 3 showing the highest risk score (34)** followed by Analyst 1 (27). This suggests potential variability in documentation practices or training levels among analysts.

**5.Majority of Records Approved but Some Review Backlog Exists**

While 75% of records are approved, **25% remain either pending or rejected**, indicating ongoing review backlog and potential delays in QC documentation approval workflows.

**6. Data Integrity Issues Across Multiple Product Categories**

Data integrity issues are distributed across several product types, suggesting that documentation risks are not limited to a single product workflow but occur across the QC testing process.

## Key Insights

- A significant number of QC records required **manual corrections**, indicating reliance on manual documentation.
- Missing data entries were observed in several records, highlighting **gaps in documentation completeness.**
- Data integrity issues were **not evenly distributed across analysts**, suggesting inconsistent documentation practices.

## Data Integrity Risk Assessment

The analysis indicates **moderate data integrity risk** associated with manual documentation processes in the QC workflow.
Risk indicators identified:

- Missing data entries
- High frequency of corrections
- Uneven risk distribution among analysts

These risks may impact regulatory compliance if not addressed.

# Recommendations

1. **Strengthen Documentation Controls**
   Introduce standardized documentation procedures to reduce missing data.
2. **Improve Data Integrity Awareness**
   Provide training programs on **ALCOA+ principles** for QC analysts.
3. **Implement Data Validation Checks**
   Introduce automated validation rules to minimize manual corrections.

# Business Impact

Strengthening data integrity controls within QC laboratories can improve:

- Documentation accuracy
- Enhance inspection readiness
- Compliance with ALCOA+ principles

Expected benefits:

- Improved data reliability
- Reduced documentation errors
- Strengthened regulatory compliance